Treatment Comparisons

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Example

- Studying the effects of 3 (isocaloric) diets on cholesterol (low-density lipoprotein, LDL):
  - "Typical" diet (control, for comparison) - μ_1
  - High fat diet - μ_2
  - High fat diet, focus on unsaturated fats - μ_3
- Questions you may ask:
  - How does each high fat diet individually compare to the typical diet? Treatment v.s. Control
    - μ_1 vs μ_2 ; μ_1 vs μ_3
  - How do the two high fat diets compare to each other? Pairwise.
    - μ_2 vs μ_3
  - How does a high fat diet in general compare to the typical diet? Difference of Averages
    Average effect of u_2
    - μ_1 vs $(\mu_2 + \mu_3)/2$

Linear Combinations of Means

- We can write the corresponding null hypotheses more generally as:

Linear

H_0: L = $\sum c_i \mu_i = L_0$, where the c_i and L_0 are constants

- How does each high fat diet individually compare to the typical diet?
  - $H_0: \mu_1 - \mu_2 = 0$; $H_0: \mu_1 - \mu_3 = 0$
- How do the two high fat diets compare to each other?
  - $H_0: \mu_2 - \mu_3 = 0$
- How does a high fat diet in general compare to the typical diet?
  - $H_0: \mu_1 - \frac{1}{2}(\mu_2 + \mu_3) = 0$

Linear Combinations of Means

- Our most common hypotheses can be written as linear combinations of means:
  - Treatment vs. control
  - Other pairwise comparisons
  - Comparing combinations of treatments
- How about $H_0: \mu_1 - \mu_2 = 10$? Ho doesn't always have to be 0.
- Planned vs. post hoc hypotheses – the distinction matters especially for controlling Type I errors

Linear Combinations of Means

- Remember:

var(aX+bY) = a^2var(X) + b^2var (Y), when X and Y are independent (like our μ_i)

ANOVA:

var(μ_i) = $\frac{\sigma^2}{n_i}$

MSE estimates σ^2, σ^2 is the common σ^2.

var($\bar{y}_{i}.$) = $\frac{MSE}{n_i}$

Linear Combinations of Means

Ex): Compare group 1 to group 2: $\hat{L} = \bar{y}_1 - \bar{y}_2$

$\Downarrow$

$c_1 = 1$, $c_2 = -1$

Coefficient

$$\hat{L} = \sum c_i \bar{y}_{i\cdot}$$ Estimate of Linear Combination

$$var(\hat{L}) = var\left(\sum c_i \bar{y}_{i\cdot}\right)$$

$$= \sum c_i^2 \, var(\bar{y}_{i\cdot})$$

$$= \sum c_i^2 \frac{MSE}{n_i}$$

$$= \boxed{MSE \sum \frac{c_i^2}{n_i}}$$

Linear Combinations of Means

$$\boxed{t_0 = \frac{\hat{L} - L_0}{\sqrt{var(\hat{L})}}}$$

Most often $L_0 = 0$

t^*

Under H$_0$: $t_0 \sim t_{N-a}$

(STAT514)

Common Types of Contrast

Pairwise Comparisons every possible $u_i - u_j$

A pairwise comparison is a difference between any two treatment effects $\tau_i - \tau_j$. The contrast coefficient is $(0, 0, \ldots, 1, 0, \ldots, -1, 0, \ldots)$ where the 1 and -1 are in the ith and jth position, respectively.

Q: If there are $v = 4$ treatments, how many pairwise comparisons that need to be investigated?

Treatment Versus Control compare every treatment to baseline

baseline

If treatment 1 is control, then the treatment versus control contrasts are $\tau_2 - \tau_1, , \ldots, \tau_v - \tau_1$.

Difference of Averages $\frac{u_1+u_4}{2} - \frac{u_3+u_4}{2}$

When the treatments divide naturally into two or more groups and the experimenter is interested in the difference of averages, the difference of averages contrasts are used. For example, in the pedestrian light experiment, it is of interest to use the contrast $(\tau_2 + \tau_3 + \tau_4)/3 - \tau_1$

Trend Contrast when treatments level are ordered (e.g. dose levels)

Used when the levels of the treatment factor are quantitative and have a natural ordering as in the hear-lung experiment. Table A.2 in the textbook lists contrast coefficients for orthogonal polynomial trend contrasts when the levels are equally spaced and the replicates are equal.

$v = 3$

Trend	c_1	c_2	c_3
Linear	-1	0	1
Quadratic	1	-2	1

$v = 4$

Trend	c_1	c_2	c_3	c_4
Linear	-3	-1	1	3
Quadratic	1	-1	-1	1
Cubic	-1	3	-3	1

Contrasts

- Linear combinations with coefficients that sum to zero
- L = $\sum c_i \mu_i$ = 0, where $\sum c_i = 0$
- How does each high fat diet individually compare to the typical diet?
  - H_0: μ_1 - $\mu_2 = 0$; H_0: μ_1 - $\mu_3 = 0$
  - c = {1, -1, 0} ; c = {1, 0, -1}

u_3 u_3

$|+0-| = 0$

Must add up to 0.

- How do the two high fat diets compare to each other?
  - H_0: μ_2 - $\mu_3 = 0$
  - c = {0, 1, -1}

u_1 u_2 u_3

- How does a high fat diet in general compare to the typical diet?
  - H_0: μ_1 - $\frac{1}{2}$(μ_2 + μ_3) = 0 $\longrightarrow u_1 = \frac{u_2 + u_3}{2}$
  - c = {1, -0.5, -0.5}
  - c = {2, -1, -1}

$u_1 = 0.5u_2 + 0.5u_3$

$u_1 - 0.5u_2 - 0.5u_3 = 0$

$\{1, -0.5, -0.5\}$

$2u_1 = u_3 + u_3$ Mathimatically correct, but:

- Must clarify interpretation of relationships

Contrasts

- Estimate L with $\hat{L}$

$$\hat{L} = \sum c_i \bar{y}_{i.}$$

- Under H_0:

$$t_0 = \frac{\hat{L}}{\sqrt{var(\hat{L})}} \sim t_{N-a}$$

- $t_{N-a}^2 = F_{1,N-a}$, so F test works too

“anova3.sas”

$$y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon_i$$

$$\varepsilon_i \sim iid\ N(0, \sigma^2)$$

<u>Trt 1</u> $X_{1j} = \begin{cases} 1 : \text{Trt } 1 \\ 0 : \text{Otherwise} \end{cases}$ $y_i = \beta_0 + \beta_1 X_{1j} + \varepsilon_i$

<u>Trt 2</u> $X_{2j} = \begin{cases} 1 : \text{Trt } 2 \\ 0 : \text{Otherwise} \end{cases}$ $y_i = \beta_0 + \beta_2 X_{2j} + \varepsilon_i$

<u>Trt 3</u> $X_{1j} = 0$

$X_{2j} = 0$ $y = \beta_0 + \varepsilon_i$

Controlling Type I Error

- The overall F test captures all possible alternatives
  - Fixes type I error rate at α
  - It's a good practice to look at overall F test first, then look at specific hypotheses only if the F test is significant ("protected F test")

    ↓ only test group comparisons if the F-test is significant (protected strategy)
- Issue when you have multiple hypotheses to test:
  - Multiple tests inflate overall type I error rate ↑ false positive
  - Not all of these tests are independent
  - It's therefore hard to know what your overall type I error risk is

Comparison of Means

- You may have several statistical tests to do
- Each one carries a risk of Type I error
- Together, your chance of at least one Type I error increases
- You could try to reduce α for each statistical test, to control the overall α
  - But remember, as α gets smaller, your power does too – tradeoff $\downarrow\alpha$, $\downarrow$ Power
- This is called the multiple comparison problem, and there is no "best" solution

Comparison of Means

- Several methods exist, and your choice depends on the set of hypotheses you want to test:
  - A few specific hypotheses?
  - Each treatment vs. the control?
  - All pairwise comparisons?
  - All possible contrasts?

🔷 **5. Multiple Comparisons**

When doing **many contrasts**, adjust for Type I error.

Methods:

Method	Best for	Notes
Bonferroni	≤10 pre-planned contrasts	Widens CI
Scheffé	All possible contrasts	Very conservative
Tukey	All pairwise	Best for pairwise
Dunnett	Treatment vs control	Focused

- Comparison methods vary in protection:
  - Experimentwise / overall Type I error
  - Individual Type I error

Any Contrast (unplanned contrasts)

F Dist.

Scheffé's method

Most flexible

- Test any linear contrast (planned or unplanned)

- Covers all possible contrasts
- Protects for unplanned comparisons
- Experimentwise type I error rate is at most α
- Results in some loss of power
- Compare absolute value of estimated contrast to a calculated value

Most Conservative

Any (few planned)

t Dist.

Bonferroni method

• Control the probability of at least one Type I error among 'm' tests.

P(at least one type I error in m tests)

$\leq m\alpha'$

So, for each test, use $\alpha' = \alpha/m$ Adjust α by dividing by # of tests

- Use for planned comparisons only
- Extremely conservative if m is large

Overly

Pairwise Comparisons of Means

- These are contrasts – a subset of all possible contrasts
- Tradeoff between power and probability of Type I error, so Scheffé may be overkill

Least significant difference (LSD)

- The usual t-test
- Controls individual but not experimentwise type I error
- Calculate:

$$\mathrm{LSD} = t_{\alpha/2,N-a}\sqrt{MS_E\left(\frac{1}{n_i}+\frac{1}{n_j}\right)}$$

- Compare:

$$|\bar{y}_{i.} - \bar{y}_{j.}| > \mathrm{LSD}$$

- Protected LSD: Only do LSD comparisons if overall F test is significant (generally a good policy)

Studentized Range Dist.

All Pairs

Tukey's Method

More powerful than Scheffé for pairwise.

• Adjust for all pairwise comparisons among group means.

- When you are interested in all possible pairwise comparisons
    - Accounts for you looking at any such pair
- Controls experimentwise error rate at α
- Uses a studentized range distribution, rather than t distribution
- Reject if:

Q > t

usually t-distribution, but this time its a "Q-distribution."

$$|\overline{y}_{i.} - \overline{y}_{j.}| > \frac{q_{\alpha,a,N-a}}{\sqrt{2}}\sqrt{\mathrm{MS_E}\left(\frac{1}{\mathrm{n_i}} + \frac{1}{\mathrm{n_j}}\right)}$$

Dunnet's Dist.

Trt v.s. Control is a type of pairwise

Dunnett's Test

More powerful than Tukey for Trt v.s. Control

- Compare each treatment to a control.

- Designed specifically for comparing individual treatments to a control (i.e., not all pairwise comparisons)
- Similar to Tukey's method
- Controls experimentwise error rate

“multiple comparisons.sas”

- There are many other methods
- The methods we’ve discussed adjust the multiplier to the SE
  - for each individual test to control FWER (Type I)
  - Alter α level (e.g., Bonferroni)
  - Use a different distribution (e.g., Tukey)
    Dunnet
- Balance between
  - Being conservative: reduce overall type I error rate / avoid false positives trade off: ↑ False Negative (Type II)
  - Being (statistically) powerful: avoid false negatives

Other Approaches

- These methods are used $\boxed{\text{after ANOVA}}$ when we want to compare group means.
- When multiple hypothesis are run, the chance of making it Type I error ↑↑↑.
- Each method controls the FWER at a specific level "α".

- Several methods we've discussed try to control the **experimentwise error rate** (also known as the **familywise error rate or FWER**)
  - Controls the probability of having at least one false positive
- Some applications (especially "big data" applications) try to control the **false discovery rate (FDR)**: the expected proportion of rejected null hypotheses that are false positives

  "How many of those rejections are false rejections?"

  - Methods to control FDR, such as the Benjamini–Hochberg (BH) or Benjamini–Yekutieli procedures, are often based on p-values (PROC MULTTEST in SAS)
  - See the last chapter of: https://www.statlearning.com/